DDD-T-86G
October 26, 1977
SUPERSEDING
Fed. Spec. DDD-T-86F
January 24, 1973

FEDERAL SPECIFICATION

TAPE, TEXTILE; COTTON, GENERAL PURPOSE

(UNBLEACHED, BLEACHED, OR DYED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cotton tape for general use.

1.2 Classification.

1.2.1 Types and classes. The cotton tape shall be of the following types and classes as specified (see 6.2).

Type I	-	Stay bindings, herringbone twill weave.
Type II	-	Other bindings, herringbone twill weave.
Type III	-	Bindings, plain weave.
Type IV	-	Binding, herringbone twill weave - special use.
Class 1	-	Unbleached.
Class 2	-	Bleached.
Class 3	-	Dyed

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standards:

FED-STD-4	-	Glossary of Fabric Imperfections.
FED-STD-123	-	Marking for Shipment (Civil agencies).
FED-STD-191	-	Textile Test Methods.

FSC 8315

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

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Military Specification:

MIL-W-43334 - Webbing and Tape, Textile, Packaging and Packing of.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with the specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Government and supply purchases. The requirements specified in 3.9 and 3.11 apply only to tape purchased directly by the Government. All other requirements apply to tape purchased as a component for an end item by a supplier and to tape purchased directly by the Government.

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3.2 Standard sample. When a standard sample is established, the tape shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Materials.

3.3.1 Yarn. The cotton yarn counts for tape shall be as specified in 3.5. A tolerance of plus or minus 5 percent shall be allowed for the specified yarn count.

3.4 Color. The tape shall be unbleached, bleached, or dyed as specified (see 6.2).

3.4.1 Labile sulfur. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished tape shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.3. When a standard sample is not available the dyed and finished tape shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.3.

3.4.2 Dyeing. The colored tape shall be stock, yarn, or piece dyed.

3.4.3 Matching. The color shall match the standard sample under artificial daylight having a color temperature of 7500 kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800 kelvin.

3.4.4 Colorfastness. Colorfastness shall be as specified in the applicable end item specification or as set forth in the contract or order (see 6.2). When a standard sample has been established, the test specimen shall be compared with the standard and shall be equal to or better than the standard for the colorfastness properties specified when tested as specified in 4.3. When no standard sample has been established, the colored tape shall be rated in accordance with the adjective ratings shown in the applicable test methods.

3.4.4.1 Standard colorfastness. In the absence of colorfastness requirements elsewhere, colored tape shall show colorfastness to laundering (after 3 cycles), light after 40 standard fading hours, and crocking equal to or better than the standard sample (see 6.4) when tested as specified in 4.3. When no standard sample has been established, the colored tape shall show "good" fastness to laundering (after 3 cycles) and light after 40 standard fading hours, and no more crocking than Munsell value 8.5 when tested as specified in 4.3. When black is specified, the tape shall show no more crocking than Munsell value 6.5 when tested for wet crocking.

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3.5 Physical requirements. The physical requirements of the tape shall conform to the requirements as specified in table I and table II when tested as specified in 4.3.

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TABLE I. Physical requirements (shuttle loom)

Type	Width, in (mm)	Tolerance, in (mm)	Warp total yarns min.	Warp yarn counts & ply cotton count (tex) ± 5.0 percent	Filling yarns per inch min. (1 cm)	Filling yarn counts & ply cotton count (tex) ± 5.0 percent	Warp strength pounds min. (newton)
I	1/4 (6)	± 1/32 (± 1)	28	40/2 or 20/1 (14.8X2 or 29.5X1)	38 (15)	60/2 or 30/1 (9.8X2 or 19.7X1)	25 (111)
I	3/8 (10)	± 1/32 (± 1)	36	40/2 or 20/1 (14.8X2 or 29.5X1)	38 (15)	60/2 or 30/1 (9.8X2 or 19.7X1)	30 (134)
I	7/16 (11)	± 1/32 (± 1)	44	40/2 or 20/1 (14.8X2 or 29.5X1)	38 (15)	60/2 or 30/1 (9.8X2 or 19.7X1)	35 (156)
I	1/2 (13)	± 1/32 ± 1)	52	40/2 or 20/1 (14.8X2 or 29.5X1)	38 (15)	60/2 or 30/1 (9.8X2 or 19.7X1)	42 (187)
I	17/32 (14)	± 1/32 ± 1)	52	40/2 or 20/1 (14.8X2 or 29.5X1)	40 (16)	20/2 (29.5X2)	42 (187)
I	5/8 (16)	± 1/32 (± 1)	60	40/2 or 20/1 (14.8X2 or 29.5X1)	38 (15)	60/2 or 30/1 (9.8X2 or 19.7X1)	50 (222)
I	11/16 (18)	± 1/32 (± 1)	68	40/2 or 20/1 (14.8X2 or 29.5X1)	38 (15)	60/2 or 30/1 (9.8X2 or 19.7X1)	58 (258)
I	3/4 (19)	± 1/32 (± 1)	76	40/2 or 20/1 (14.8X2 or 29.5X1)	38 (15)	60/2 or 30/1 (9.8X2 or 19.7X1)	65 (289)
I	13/16 (21)	± 1/32 (± 1)	84	40/2 or 20/1 (14.8X2 or 29.5X1)	38 (15)	60/2 or 30/1 (9.8X2 or 19.7X1)	72 (320)
I	7/8 (22)	± 1/32 (± 1)	92	40/2 or 20/1 (14.8X2 or 29.5X1)	38 (15)	60/2 or 30/1 (9.8X2 or 19.7X1)	80 (356)
I	1 (25)	± 1/32 (± 1)	100	40/2 or 20/1 (14.8X2 or 29.5X1)	38 (15)	60/2 or 30/1 (9.8X2 or 19.7X1)	85 (378)
I	1-1/16 (27)	+ 1/16 - 1/32 (+ 2) (-1)	108	40/2 or 20/1 (14.8X2 or 29.5X1)	38 (15)	60/2 or 30/1 (9.8X2 or 19.7X1)	90 (400)
I	1-1/8 (29)	+ 1/16 - 1/32 (+ 2) (-1)	116	40/2 or 20/1 (14.8X2 or 29.5X1)	38 (15)	60/2 or 30/1 (9.8X2 or 19.7X1)	95 (423)
I	1-1/4 (32)	+ 1/16 - 1/32 (+ 2) (-1)	132	40/2 or 20/1 (14.8X2 or 29.5X1)	38 (15)	60/2 or 30/1 (9.8X2 or 19.7X1)	115 (512)
I	1-1/2 (38)	+ 1/16 - 1/32 (+ 2) (-1)	164	40/2 or 20/1 (14.8X2 or 29.5X1)	38 (15)	60/2 or 30/1 (9.8X2 or 19.7X1)	130 (578)

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TABLE I. Physical requirements (shuttle loom) (cont'd)

Type	Width, in (mm)	Tolerance, in (mm)	Warp total yarns min.	Warp yarn counts & ply cotton count (tex) ± 5.0 percent	Filling yarns per inch min. (1 cm)	Filling yarn counts & ply cotton count (tex) ± 5.0 percent	Warp strength pounds min. (newton)
II	3/16 (5)	± 1/32 (± 1)	32 1/	60/2 (9.8X2)	58 (23)	60/2 (9.8X2)	22 (98)
II	1/2 (13)	± 1/32 (± 1)	74 2/	60/2 (9.8X2)	70 (28)	30/1 or 60/2 (19.7X1 or 9.8X2)	50 (222)
II	5/8 (16)	± 1/32 (± 1)	94 3/	60/2 (9.8X2)	70 (28)	30/1 or 60/2 (19.7X1 or 9.8X2)	60 (267)
III	3/16 (5)	± 1/32 (± 1)	13	24/2 (24.6X2)	28 (11)	30/1 or 60/2 (19.7X1 or 9.8X2)	14 (62)
III	1/4 (6)	± 1/32 (± 1)	17	24/2 (24.6X2)	28 (11)	30/1 or 60/2 (19.7X1 or 9.8X2)	18 (80)
III	3/8 (10)	± 1/32 (± 1)	25	24/2 (24.6X2)	28 (11)	30/1 or 60/2 (19.7X1 or 9.8X2)	24 (107)
III	1/2 (13)	± 1/32 (± 1)	34	24/2 (24.6X2)	28 (11)	30/1 or 60/2 (19.7X1 or 9.8X2)	35 (156)
III	9/16 (14)	± 1/32 (± 1)	35	24/2 (24.6X2)	28 (11)	30/1 or 60/2 (19.7X1 or 9.8X2)	36 (160)
III	1 (25)	± 1/32 (± 1)	65	24/2 (24.6X2)	28 (11)	30/1 or 60/2 (19.7X1 or 9.8X2)	65 (289)
IV	1-1/4 (32)	± 1/32 (± 1)	132	20/2 (29.5X2)	46 (18)	40/2 (14.8X2)	185 (823)

1/ 8 ends left - 16 ends right - 8 ends left.

2/ 20 ends left - 34 ends right - 20 ends left.

3/ 22 ends left - 50 ends right - 22 ends left.

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TABLE II. Physical requirements (shuttleless loom)

Type	Width, in (mm)	Tolerance, in (mm)	Warp total yarns min.	Warp yarn counts & ply cotton count (tex) ± 5.0 percent	Filling yarns per inch min. (1 cm)	Filling Yarn counts & ply cotton count (tex) ± 5.0 percent	Warp strength pounds min. (newton)	Catch-cord yarn size cotton count (tex)
I	3/4 (19)	± 1/32 (± 1)	76	40/2 or 20/1 (14.8X2 or 29.5X1)	76 1/ (30)	60/1 (9.8X1)	65 (289)	60/1 (9.8/1)
III	1 (25)	± 1/32 (± 1)	65	24/2 (24.6X2)	56 1/ (22)	60/1 (9.8X1)	65 (289)	60/1 (9.8/1)

1/ 2 picks per shed.

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3.5.1 Weave.

3.5.1.1 Type I (shuttle loom). The weave shall be a two up and two down single or multiple herringbone twill with one or more reversals of the twill across the width of the tape. The tape shall have a woven edge on both sides.

3.5.1.1.1 Type I (shuttless loom). The weave shall be a two up and two down single or multiple herringbone twill with one or more reversals of the twill across the width of the tape. The filling shall traverse the full width of the tape and shall be held at the edge by an extra catch-cord end interlacing with the filling yarn in a method depicted in figure 1 or 2.

3.5.1.2 Type II (shuttle loom). The weave shall be a two up and two down herringbone twill as specified in 3.5.1.1. The tape shall have a woven edge on both sides.

3.5.1.3 Type III (shuttle loom). The weave shall be plain. The tape shall have a woven edge on both sides.

3.5.1.3.1 Type III (shuttleless loom). The weave shall be plain. The filling shall traverse the full width of the tape and shall be held at the edge by an extra catch-cord end interlacing with the filling yarn in a method depicted in figure 1 or 2.

3.5.1.4 Type IV (shuttle loom). The weave shall be a two up and two down single or multiple herringbone twill with one or more reversals of the twill across the width of the tape. The tape shall have a woven edge on both sides.

3.6 Nonfibrous material. The starch and protein content including chloroform-soluble and water-soluble material shall not exceed 10.0 percent for class 1 tape; 2.5 percent for class 2 tape; and 4.0 percent for class 3 tape, when tested as specified in 4.3.

3.7 Shrinkage. When specified (see 6.2), the classes 1, 2, and 3 tape shall be preshrunk and shall not shrink more than 4.0 percent in the direction of the warp, when tested as specified in 4.3. The preshrinking process used shall not be identified by name or trademark on the tape, ticket, label, or package.

3.8 Curvature. The finished type I and III, shuttleless loom tape, shall show no more lateral curvature than 1/4 inch (6 mm) within a yard when tested as specified in 4.3.

3.9 Put-up. The tape shall be wound on standard commercial spools containing the yardage and allowable number of pieces as specified in table III. No piece

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shall be less than 36 yards (33 m) in length. When specified (see 6.2), small requirements for tape may be put up on 36-yard (33 m) rolls in packages of twenty rolls. The number of splices in any package of twenty 36-yard (33 m) rolls shall not exceed three. No piece shall be less than 10 yards (9 m).

TABLE III. Put-up

Width inch (mm)	Spool length yds. ± 2(meters) percent	Number of pieces (max.) Type I	Type II	Type III	Type IV
3/16 (5)	720 (658)		6		
1/4 (6)	720 (658)	4			
3/8 (10)	720 (658)	4			
7/16 (11)	720 (658)	4			
1/2 (13)	720 (658)	5	5		
17/32 (14)	720 (658)	5			
5/8 (16)	720 (658)	5	5		
11/16 (18)	720 (658)	6			
3/4 (19)	720 (658)	6			
13/16 (21)	720 (658)	7			
7/8 (22)	720 (658)	7			
1 (25)	720 (658)	8			
1-1/16 (27)	504 (461)	5			
1-1/8 (29)	504 (461)	5			
1-1/4 (32)	504 (461)	5			
1-1/2 (38)	504 (461)	6			
3/16 (5)	1000 (914)			6	
1/4 (6)	1000 (914)			6	
3/8 (10)	1000 (914)			6	
1/2 (13)	1000 (914)			7	
9/16 (14)	1000 (914)			7	
1 (25)	720 (658)			9	
1-1/4 (32)	504 (461)				5

3.10 Fiber identification. Each spool or package shall be labeled or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act.

3.11 Identification label. Each spool or package of tape shall have an identification gummed label attached to the head end of the spool or to the

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outside end of the package as applicable. The color of the label shall be light in intensity to permit easy reading of printed, stamped or typed markings. The use of handwritten entries is prohibited. The label shall be legibly printed with water insoluble ink with the following information:

Stock number.
Nomenclature.
Yardage.
Contract number and date of manufacture (month and year)
Specification number.
Supplier's name.

3.12 Workmanship. The tape shall conform to the quality of product established by this specification. Occurrence of defects shall not exceed the applicable acceptable quality levels (AQLs).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.2.2 End item examination. Defects found during the examination shall be classified in accordance with 4.2.2.1 through 4.2.2.3.

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4.2.2.1 Yard-by-yard examination for visual defects. The required yardage of each spool or roll shall be inspected on both sides and visual defects classified as listed in table IV. All defects found shall be counted regardless of their proximity one to another, except where two or more defects represent a single local condition of the tape in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit for this examination shall be 1 linear yard (0.91 m). The sample size shall be in accordance with level II of MIL-STD-105. The AQL shall be 0.4 major and 1.5 total (major and minor combined) defects per 100 units (yards) (per 0.91 m). The lot size shall be expressed in units of 1 yard (0.91 m) each. An approximate equal number of yards shall be examined from each spool or roll selected.

TABLE IV. Classification of defects

Glossary No. (FED-STD-4) (Sec 1)	Defects	Classification Major	Minor
79	Abrasion marks resulting in a weak place	X	
33	Broken or missing ends-two or more within a linear inch, regardless of size of missing portion	X	
	Single, 1/2 inch (13 mm) or more in length		X
90, 91, 92	Cut, hole, or tear-any	X	
74, 75, 76	Spot, stain, or streak-clearly visible 1/		X
13, 34	Float or skips-multiple, more than 1/4 inch (6 mm) in combined warp and filling directions	X	
	Multiple, 1/4 inch (6 mm) or less in combined warp and filling directions		X
	Single, floating over 1 inch (25 mm) or more in length		X
	Single, over more than 1/2 (13 mm) the width of the tape		X
	Heavy filling bar, clearly visible 1/		X

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TABLE IV. Classification of defects (cont'd)

Glossary No. (FED-STD-4) (Sec 1)	Defects	Classification Major	Minor
3, 5, 27	Jerked-in or sloughed-off filling, slub, or slug-clearly visible 1/		X
6, 28	Kink or loop on surface of tape		X
7, 29	Knots-large, improperly pieced knots, clearly visible, 1/ protruding from surface or with tails extending from surface of tape		X
9	Light or fine filling bar-clearly visible 1/	X	
10	Mispick- any		X
49, 54	Edges frayed, slack, rolled, wavy, or otherwise poorly constructed, 4-1/2 inches (114 mm) or more	X	
	Less than 4-1/2 inches (114 mm) but more than 1/2 inch (13 mm)		X
36, 46	Mixed or dirty ends		X
78	Smash-any	X	
40	Wrong draw-9 inches (229 mm) or more in length		X

1/ Clearly visible at normal inspection distance (approximately 3 feet (1 meter).

4.2.2.2 Overall examination. Each defect listed below shall be counted no more than once in each spool or roll examined. The sample unit for this examination shall be one spool or roll. The sample size and acceptance number shall be as shown in table V.

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Defects

Off shade, not within established tolerances.
Uneven dyeing.
Poor dye penetration.
Shaded throughout.
Objectionable odor.
Overall uncleanness.
Not labeled in accordance with Textile Fiber Products Identification Act.

TABLE V. Sample size

Lot size in spools or rolls	Sample size in spools or rolls	Maximum number of defects acceptable in sample (see 4.2.2.2 and 4.2.2.3.1)
5 or less	1	0
6 up to and including 15	2	0
16 up to and including 40	3	0
41 up to and including 110	5	0
111 up to and including 300	7	0
301 and over	10	1

4.2.2.3 Length and width examination.

4.2.2.3.1 Examination for length and width of tape on individual spools or rolls. Each defect listed below shall be counted no more than once in each spool or roll examined. The sample unit for this examination shall be one spool or roll. The sample size and acceptance number shall be as shown in table V.

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Defects

Total yardage on spool or roll less than the applicable minimum or more than the applicable maximum length:

Type	Width	yds(m) Length Minimum	yds(m) Maximum
I and II	1 inch (25 mm) or less	706(646)	734(671)
I	More than 1 inch (25 mm)	494(452)	514(470)
III	3/16 (5 mm) thru 9/16 inch (14 mm)	980(896)	1020(933)
III	1 inch (25 mm)	706(646)	734(671)
I, II, III (36 yd. (33 m) rolls x 20)	Any	720(658)	-
IV	1-1/4 inches (32 mm)	494(452)	514(470)

Width not within specified tolerance.
Any piece less than minimum specified.
More than the number of pieces of splices specified.
Actual gross length more than 2 yards (1.8 m) less than gross length marked on spool ticket.

4.2.2.3.2 Examination for total yardage in sample. The lot shall be unacceptable if the total of the actual gross lengths of spools or rolls in the sample is more than 5 yards (4.6 m) less than the total of the gross lengths marked on spool or package tickets.

4.2.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of MIL-W-43334 to determine that packaging, packing, and marking comply with the section 5 requirements.

4.3 Tests. The method of testing specified in FED-STD-191, however applicable, and as listed in table VI, shall be followed. The physical and chemical values specified in section 3 apply to the results of the

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determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The sample size shall be as follows:

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more units fail to meet any requirement specified. The lot size shall be expressed in units of 1 linear yard (0.91 m). The sample unit for testing shall be as follows:

1/4 inch (6 mm) thru 3/8 inch (10 mm) in width - 25 linear yards (23 m)
7/16 inch (11 mm) 5/8 inch (16 mm) in width - 20 linear yards(18 m)
11/16 inch (18 mm) thru 7/8 inch (22 mm) in width - 15 linear yards (14 m)
1 inch (25 mm) and wider - 10 linear yards (9 m)

TABLE VI. Test methods

Characteristic	Requirement paragraph	Test method
Cotton yarn:		
Material	3.3.1	2/
Count	3.5	4021 2/
Single	3.5	Visual 1/
Ply	3.5	Visual 1/
Weave	3.5.1	Visual 1/
Yarns:		
Total warp yarns	3.5	5050
Filling yarns per inch	3.5	5050
Breaking strength:		
Warp	3.5	4108 3/

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TABLE VI. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Catch-cord (type I and III shuttleless tape):		
Yarn size	3.5.1.1.1 and 3.5.1.3.1	2/
Construction	3.5.1.1.1 and 3.5.1.3.1	2/
Nonfibrous material	3.6	2611
Labile sulfur	3.4.1	2020
Colorfastness to:		
Laundering (after 3 cycles)	3.4.3	5610 4/ 5/
Light	3.4.3	5660 6/
Crocking	3.4.3	5651
Wet drycleaning	3.4.3	5622
Accelerated weathering	3.4.3	5671 6/
Perspiration	3.4.3	5680
Shrinkage	3.7	4.3.1
Curvature (type I and III shuttleless tape)	3.8	4.3.2

1/ One determination per sample unit and results reported as "pass" or "fail".

2/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

3/ Either split drum or flat surface clamps may be used. When flat surface clamps are used, the pulling clamp shall have a uniform speed of 12 ± 0.5 inches (30.5 ± 1.3 cm) per minute, the distance between jaws shall be 3 inches (7.6 cm) and the test specimen shall be a minimum of 6 inches (15.2 cm).

4/ The specimens must be dried after each of the 3 laundering cycles.

5/ Only the stain on the cotton fibers of the transfer cloth shall be evaluated.

6/ Time of exposure shall be 40 standard fading hours.

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4.3.1 Shrinkage test for tape.

4.3.1.1 Preparation. Five specimens of tape, each 20 inches (501 mm) in length, shall be used for this determination, and bench marks for measurement placed 18 inches (457 mm) apart.

4.3.1.2 Procedure. Four liters of water at 100° ± 5°F (38° ± 3°C) not over 50 parts per million (p.p.m) hardness shall be entered into a suitable glass or metal container. The water shall contain 0.2 percent (based on the weight of the tape) of an anionic wetting agent. The tape specimens shall be placed into the container and kept beneath the surface of the water for a period of one-half hour without agitation. At the end of this period the specimen shall be centrifugally extracted for 5 minutes in a 17-inch (432 cm) diameter extractor at a speed of 1575 revolutions per minute (r.p.m.) or in any other extractor for a time that will produce an equivalent extraction, and dried in a circulation air oven at a temperature of 200°F (93°C). After drying, the specimens shall be pressed using a conventional heated hand iron, care being taken to prevent stretching of tape.

4.3.1.3 Measurement of shrinkage. After reaching equilibrium in the standard atmosphere, the length of the specimen shall be measured and the shrinkage computed according to the following formula:

$$\text{Shrinkage, percent} = \frac{\text{Original measurement} - \text{measurement after shrinkage}}{\text{Original measurement}} \times 100$$

Measurements shall be taken to the nearest 0.1 inch (2.5 mm) and shrinkage shall be reported to the nearest 0.1 percent.

4.3.2 Measurement of lateral curvature.

4.3.2.1 Test specimen. The test specimen shall be a length of tape, full width, measuring a minimum of 40 inches (1016 mm). The specimen shall not be stretched, smoothed or otherwise changed from its original condition prior to testing.

4.3.2.2 Number of determinations. Five specimens shall be tested from each sample unit and averaged.

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4.3.2.3 Apparatus.

Plexiglass or equal	-	plexiglass weighing approximately 35 ounces (992 g) with dimensions of 45 inches (1143 mm) by 5 inches (127 mm) by 1/4 inch (6 mm)
Straight edge	-	a rigid straight edge measuring 36 inches (914 mm) in length
Roller	-	a roller one inch (25 mm) in diameter and weighing 1-1/2 pounds (680 g)

4.3.2.4 Procedure. The specimens shall be placed flat, on a smooth, horizontal flat surface without tension and allowed to reach moisture equilibrium as defined in section 4 of FED-STD-191. After equilibrium is reached, a weight shall be placed at one end of the tape. The roller shall be placed on the specimen at the end of the tape where the weight is located. The specimen should be approximately in the center of the roller. The roller shall be rolled along the length of the specimen, care being taken to keep the specimen in the center of the roller and not to exert any pressure on the roller. When the roller has passed the length of the tape, the plexiglass shall then be placed on the specimen for a period of 1 hour.

Without moving the plexiglass on the specimen, the straight edge shall be placed on the plexiglass so that both ends of the straight edge are aligned perpendicularly with the outermost edge of the specimen. Determine the highest degree of curvature of the specimen from the straight edge by measuring to the nearest 1/32 of an inch (0.8 mm) perpendicularly from the straight edge. Record the highest measure (see figure 3).

4.3.2.5 Report. The average of five determinations from each sample unit shall be taken.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Levels A, B and C. Tape, put up as specified, shall be packaged in accordance with the applicable requirements of MIL-W-43334. When tape is put up on 36-yard (33 m) rolls and in 20 rolls per package, the quantity of rolls and length of package limitation specified in MIL-W-43334 shall be waived.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. Tape shall be packed in accordance with the applicable requirements of MIL-W-43334.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be in accordance with MIL-W-43334.

6. NOTES

6.1 Intended use. The tape is utilized as bindings in the fabrication of clothing and other textile items. Type I, 1/2 inch (13 mm) and 17/32 inch (13.5 mm) tapes are utilized as backseam tapes in footwear items.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2.1).
(c) Color required (see 3.4).
(d) Colorfastness requirements (see 3.4.4).
(e) When class 1, 2, or 3 preshrunk tape is required (see 3.7).
(f) Put-up, when small rolls are specified (see 3.8).
(g) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 A suggested but not mandatory dyestuff formulation for Olive Drab 7 is as follows:

Vat Olive T
Vat Olive Green B, Pr. No. 293 or Vat Olive GGL
Vat Khaki 2G, Pr. No. 122

Shaded with either or both of the following:

Vat Brown R.C.I. No. 1151 or Vat Brown BR, Pr. No. 118
Vat Yellow of suitable fastness

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6.4 For access to the standard sample, address procuring activity issuing the invitation for bids.

Custodians:

Army - GL
Navy - SA
Air Force - 99

Review activities:

Army - MD, EA
Navy - MC
DLA - CT

*User activities:

Army - ME, WC
Navy - SH
DLA - DM

Preparing activity:

Army - GL

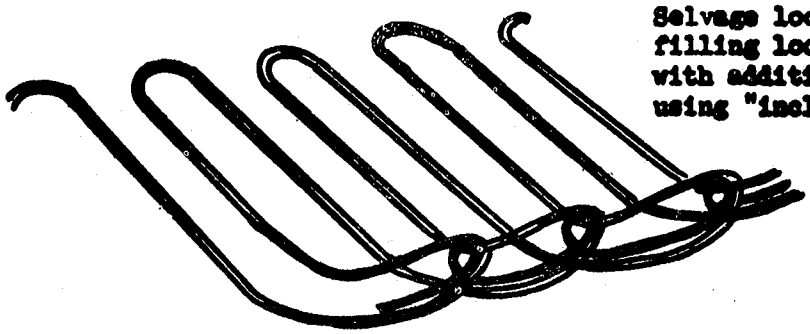
Civil Agency Coordinating activities:

GSA - FSS
HEW - HSM
VA - DMS
NASA - JFK

Project No. 8315-0239

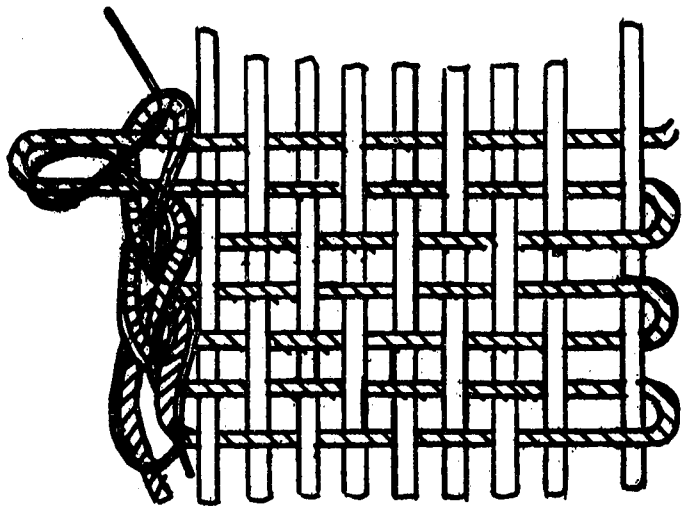
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Figure 1
Catch-Cord Diagram



Selvage looked by knitting filling loops simultaneously with additional catch thread using "inclined" latch needle.

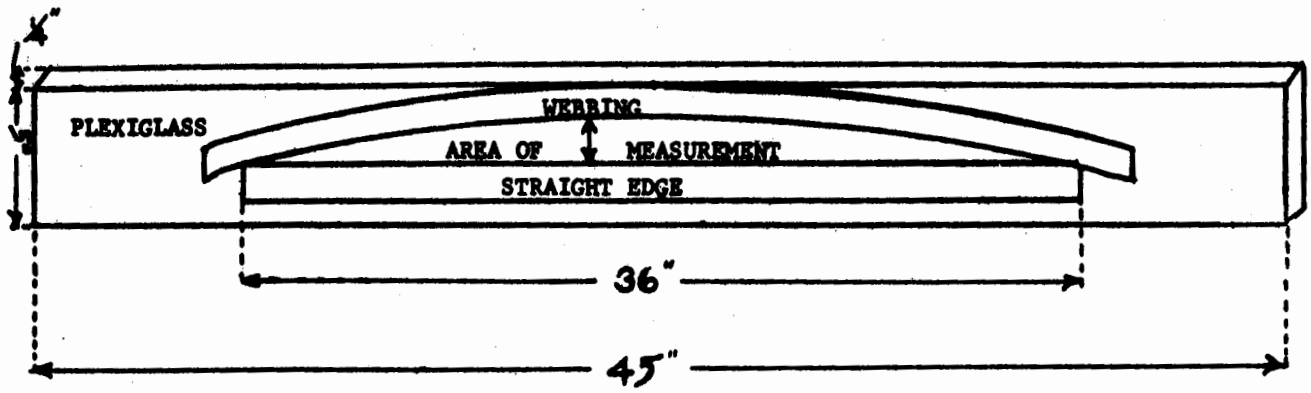
Figure 2
Catch-Cord Diagram



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Figure 3

Diagram Curvature Measurement



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